Takku Li

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EDUCATION

Northeastern University **Sep 2023 - Dec 2025**
Master of Science, Computer Science
- **Coursework:** Object-Oriented Design, Distributed Systems, Database, Cloud Computing, Networking, Robotics, Operation System

TECHNICAL SKILLS

- **Languages**: Python, Java, Triton, Golang, C/C++, JavaScript, SQL, TypeScript, Scala
- **Databases & OS**: MySQL, Redis, DynamoDB, MongoDB, PostgreSQL, Cassandra, Elasticsearch, Linux
- **Framework & Libraries**: PyTorch, Flask, Spring Boot, Express, Node.js, React, Angular, FastAPI, Django, Dask
- **Cloud**: AWS, Azure, CI/CD, GitHub Actions, Docker, Git, Microservices
- **Messaging & Stream**: Kafka, Kubernetes, RabbitMQ, ZeroMQ, Apache Pulsar, gRPC, WebSockets, GraphQL
- **Tools**: Terraform, JUnit, Postman, Spark, Maven, RESTful APIs, Nginx, Jenkins, Claude code, ChatGPT, Copilot, Gemini

EXPERIENCE

Noetic Contemplation **Jan 2026 - Present**
AI Founding Engineer *Remote*
- Built a personalized AI novel generation platform from 0→1, designing system architecture and LLM pipelines.
- Developed agent memory architecture combining short-term context and long-term retrieval, enabling seamless story continuation across user sessions and improving engagement.
- Engineered optimized generation pipelines for low-latency and cost-efficient inference, ensuring scalable, reliable delivery of content.

Cogitive **Jan 2026 - Present**
Machine Learning Engineer *San Mateo, CA*
- Building and maintaining end-to-end ML infra, including data pipelines, evaluation workflows, and model deployment systems.
- Collaborating with engineers and machine learning research scientists on research, analysis, and prototype development.

Berkshire Hathaway HomeServices **Jun 2024 - Aug 2024**
Machine Learning Engineer Intern *Seattle, WA*
- Productionized a prediction system, reducing error by 30% through optimized inference and robust backend integration.
- Designed RESTful APIs with Flask for model inference, integrating with existing applications through service mesh.
- Created CI/CD pipeline with GitHub Actions, enabling continuous testing and model deployment capability.
- Integrated AWS services (S3, IAM, CloudWatch) for data storage, access control, and real-time monitoring, ensuring high reliability and system transparency.
- Applied Agile Scrum practices on data pipeline development, reducing cycle time by 25% through automation.

Temple University **Sep 2021 - May 2023**
PhD Research Assistant in information system and quantitative marketing *Philadelphia, PA*
- Examined the influence of streaming comments on video appreciation through a data-driven research project in Python and Dask.
- Conducted a high-performance NLP pipeline by combining fine-tuned transformer models for topic classification and sentiment analysis, achieving scalable and accurate text understanding, demonstrating proficiency in advanced AI/ML libraries.

PROJECTS

Language-Conditioned VLA Robotics System
- Engineered a production-style Vision-Language-Action (VLA) robotics stack using ROS 2, combining LLM-based instruction understanding, visual perception, and Nav2 navigation for end-to-end language-conditioned robot behavior.
- Built distributed multimodal inference pipelines with multi-threaded C++/Python and DDS networking (TCP/UDP), enabling low-latency communication across perception, planning, and control modules.

AI-Driven Flashcard Generator | Qualcomm & Microsoft AI Hackathon
- Built flashcard generator to help users efficiently learn and retain new knowledge in edge devices with FastAPI, leveraging LLMs for context-awareness within an MVC backend.
- Automated over 70% of codebase generation using Vibe Coding on Claude Code with agent and skill workflows, leveraging structured prompts and enforcing MVVM architectural patterns to accelerate development while maintaining reliability and scalability.

Async Task Scheduler
- Architected distributed job scheduling framework with Go, implementing producer-consumer pattern for scalability.
- Optimized database schema with MySQL sharding and normalization, reducing query latency by 40%.
- Integrated Redis caching for hot read offloading, achieving 5x system throughput gain from 2K to 10K QPS.